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LAB SESSION 01

Project Selection and Team Formation

Objective

1. Form a student team
2. Select a project theme
3. Understand the problem
4. Setup Agile task board

Basic Theory

Agile is a method where software is built step by step.
Work is divided into small tasks and tracked using a board.

Requirement gathering means understanding what the user wants from the system.

Lab Activities

1. Form a team of 3 or 4 students
2. Assign roles
   a. Product Owner
   b. Scrum Master
   c. Developer(s)
3. One student acts as client
4. Other students ask questions to understand the problem
5. Create a task board using Trello, Jira, or GitHub
6. Create four columns
   a. Backlog
   b. In Progress
   c. Testing
   d. Done

Deliverables

1. Team member list with roles
2. One page project description
3. Screenshot of task board
4. List of 10 questions asked to the client

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Example LAB SESSION 01: Project Selection and Team Formation

Theme: Smart Library Management System

1. Team Member List & Roles

- **Product Owner:** Rahul (Interface with client, defines features)
- **Scrum Master:** Priya (Manages the Trello board, removes blockers)
- **Developers:** Amit & Sarah (Coding and Testing)

2. Project Abstract

"The Smart Library Management System is a web-based application designed to automate the borrowing and returning of books. Currently, the college uses a manual register which causes errors and delays. Our system will allow students to search for books online, reserve them, and track their due dates. It will also calculate fines automatically for late returns."

3. Questionnaire (Asked to Client)

1. Who will be the primary users of the system (Students, Faculty, Staff)?
2. Should students be able to reserve a book that is currently checked out?
3. What is the maximum number of books a student can borrow at once?
4. How are late fees calculated (per day or fixed amount)?
5. Do you need an email notification system for overdue books?
6. Should the system generate monthly reports on book usage?
7. Is there a need for barcode scanning integration?
8. Do we need to migrate old data from the manual registers?
9. What security measures are needed for student data?
10. Does the system need to run on mobile devices?

4. Task Board Screenshot (Simulation)

- **Backlog:** User Login, Search Book, Reserve Book, Calculate Fine.
- **In Progress:** Setup Database.
- **Testing:** None.
- **Done:** Project Abstract.

LAB SESSION 02

Requirement Analysis

Objective

1. Identify system requirements
2. Write simple user stories
3. Draw use case diagram

Basic Theory

Functional requirements describe what the system should do.
Non-functional requirements describe how the system should work.

A user story is written as:
As a user, I want some feature, so that I get some benefit.

Lab Activities

1. Write at least
   a. 10 functional requirements
   b. 5 non-functional requirements
2. Convert functional requirements into user stories
3. Add user stories to backlog
4. Draw use case diagram showing users and actions

Deliverables

1. Requirement document
2. Product backlog with user stories
3. Use case diagram

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Example LAB SESSION 02: Requirement Analysis

1. Functional Requirements (FR)

1. The system shall allow students to login using their university ID.
2. The system shall allow users to search for books by Title, Author, or ISBN.
3. The system shall allow librarians to add new book records.
4. The system shall calculate fines: $1 per day after the due date.
5. The system shall prevent borrowing if the user has an unpaid fine.
6. The system shall send an email reminder 2 days before the due date.
7. The system shall allow users to view their borrowing history.
8. The system shall allow a user to cancel a reservation.
9. The system shall update book status to "Available" upon return.
10. The system shall allow the admin to block a user account.

2. Non-Functional Requirements (NFR)

1. **Performance:** Search results must load within 2 seconds.
2. **Security:** Passwords must be stored using SHA-256 encryption.
3. **Availability:** The system should be available 99.9% of the time during college hours.
4. **Scalability:** The database must support up to 50,000 book records.
5. **Usability:** The UI should be responsive for mobile screens.

3. Product Backlog (User Stories)

- "As a **Student**, I want to **search for a book**, so that I can see if it is available."
- "As a **Librarian**, I want to **view overdue books**, so that I can send notices."
- "As a **Student**, I want to **renew my book online**, so that I don't get a fine."

4. Use Case Diagram

- **Actors:** Student, Librarian, Database System.
- **Use Cases:** Login, Search Book, Issue Book (Librarian only), Return Book, Pay Fine.

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LAB SESSION 03

Class Design

Objective

Design the structure of the system using classes.

Basic Theory

Classes represent real world objects.
Attributes store data.
Methods perform actions.

Lab Activities

1. Identify classes from user stories
2. List attributes and methods for each class
3. Create class responsibility cards
4. Draw class diagram with relationships

Deliverables

1. List of classes with responsibilities
2. Class diagram with minimum 6 classes

Example LAB SESSION 03: Class Design

1. List of Classes

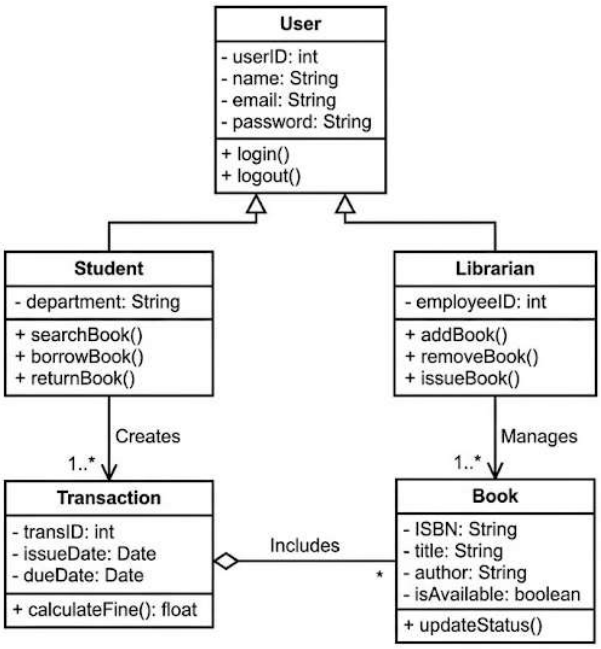
- `User` (Base class for Student/Librarian)
- `Book`
- `Transaction` (Records borrowing details)
- `Fine`
- `Catalog` (Search interface)

2. Class Responsibilities (CRC Approach)

- **Class:** `Book`
  - **Attributes:** `ISBN`, `Title`, `Author`, `Status` (Available/Issued)
  - **Methods:** `updateStatus()`, `getDetails()`
- **Class:** `Transaction`
  - **Attributes:** `TransID`, `DateOfIssue`, `DueDate`, `ReturnDate`
  - **Methods:** `calculateLateDays()`, `closeTransaction()`

3. Class Diagram Output

- **Inheritance:** `Student` and `Librarian` inherit from `User`.
- **Association:** `Student` has a "borrows" relationship with `Book`.
- **Composition:** `Library` is composed of a `Catalog`.



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LAB SESSION 04

Interaction Design

Objective

Show how different parts of the system interact.

Basic Theory

Sequence diagram shows step-by-step interaction.
Activity diagram shows workflow of the system.

Lab Activities

1. Select two important use cases
2. Draw sequence diagram for each use case
3. Draw one activity diagram for full system flow

Deliverables

1. Two sequence diagrams
2. One activity diagram

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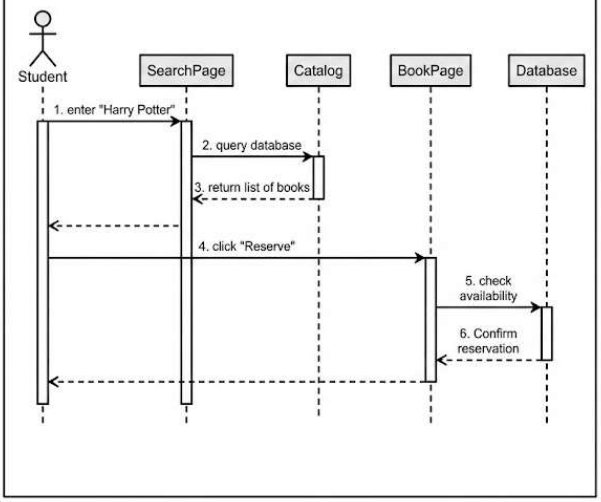
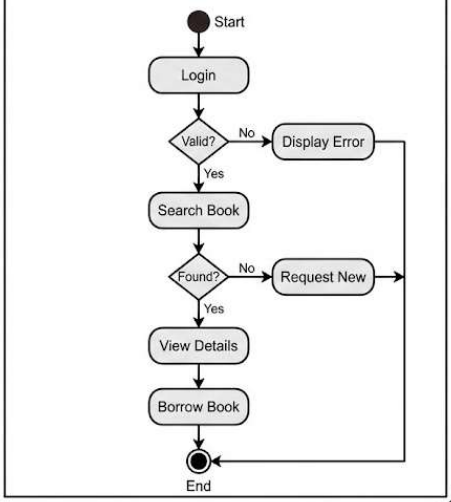
Smart Library Management System - Lab Session Diagrams

Sequence Diagram: Student Borrows a Book



Activity Diagram: Library Book Borrowing Process



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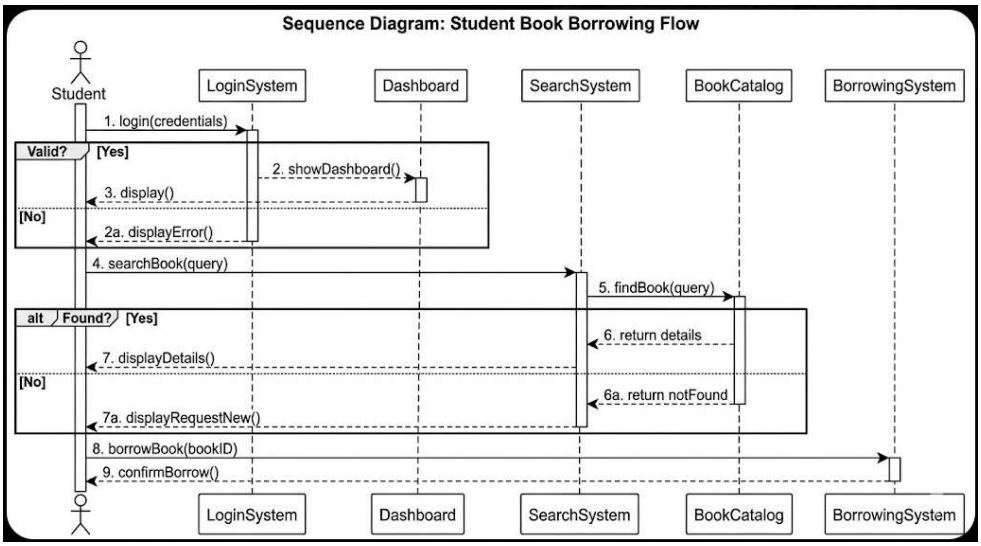
Example LAB SESSION 04: Interaction Design

1. Sequence Diagram (Scenario: Borrowing a Book)

- **Actor:** Student
- **Step 1:** Student -> `SearchPage`: enter "Harry Potter"
- **Step 2:** `SearchPage` -> `Catalog`: query database
- **Step 3:** `Catalog` -> `SearchPage`: return list of books
- **Step 4:** Student -> `BookPage`: click "Reserve"
- **Step 5:** `BookPage` -> `Database`: check availability
- **Step 6:** `Database` -> `BookPage`: Confirm reservation

2. Activity Diagram (Overall Flow)

- Start -> Login -> Valid? (No -> Error) (Yes -> Dashboard) -> Search Book -> Found? (No -> Request New) (Yes -> View Details) -> Borrow -> End.



LAB SESSION 05

Architecture Design

Objective

Understand how software components are organized and deployed.

Basic Theory

Component diagram shows software parts.
Deployment diagram shows where software runs.

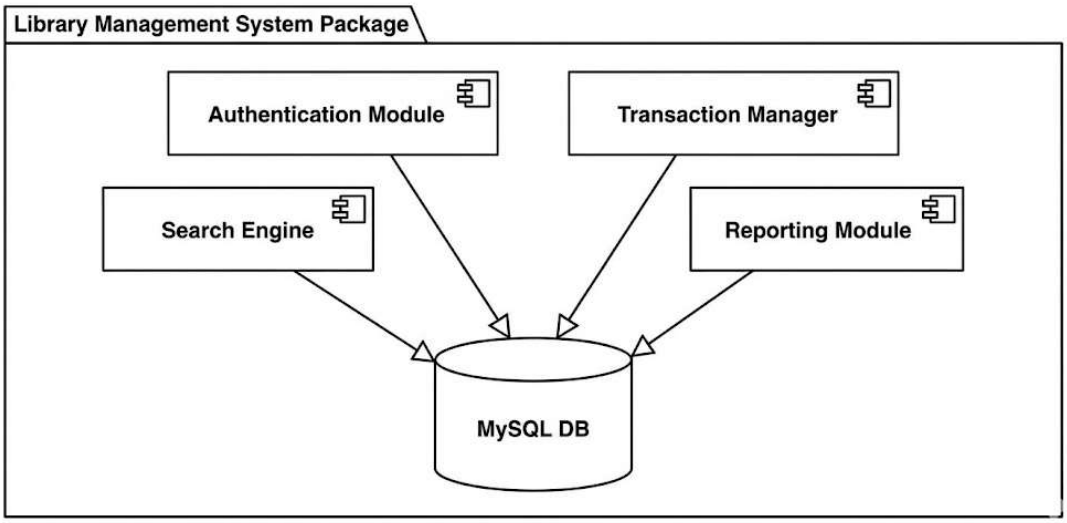
Lab Activities

1. Choose architecture type
2. Draw component diagram
3. Draw deployment diagram

Deliverables

1. Short architecture description
2. Component diagram
3. Deployment diagram

Component Diagram: Smart Library Management System



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Example LAB SESSION 05: Architecture Design

1. Architecture Type

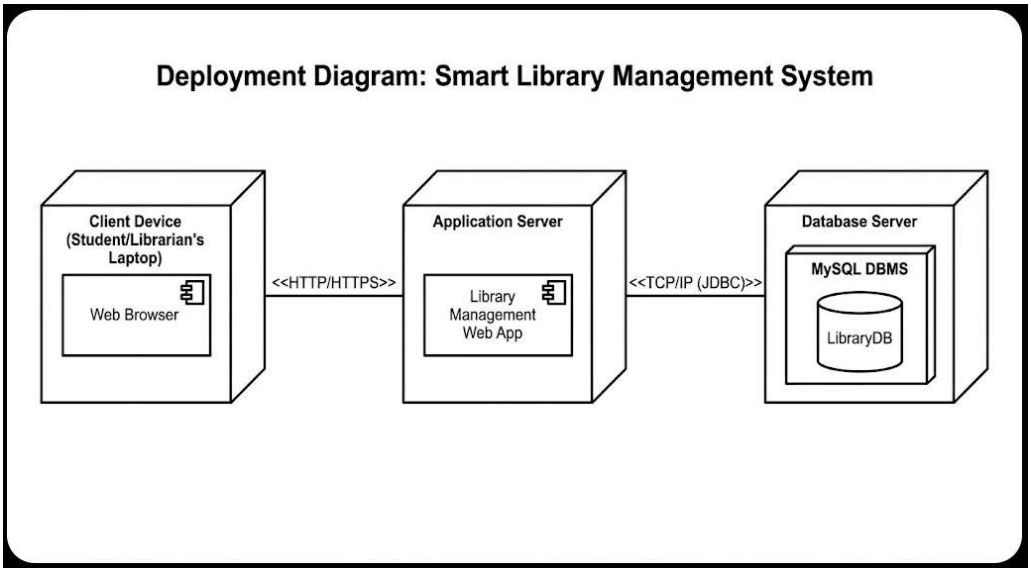
- **Model-View-Controller (MVC)** Pattern.
  - **Model:** Handles data logic (Book, Student classes).
  - **View:** The HTML/CSS pages the user sees.
  - **Controller:** Processes user input (e.g., clicking "Search") and updates the Model.

2. Component Diagram

- **Components:** `Authentication Module`, `Search Engine`, `Transaction Manager`, `Reporting Module`.
- **Database:** `MySQL DB`.

3. Deployment Diagram

- **Node 1 (Client):** Student's Laptop (Web Browser).
- **Node 2 (Server):** Application Server (Apache Tomcat/Node.js).
- **Node 3 (Database):** Database Server (MySQL).



LAB SESSION 06

Implementation – Part 1

Objective

Start coding the project.

Basic Theory

Clean code is easy to read and understand.
Git is used to manage code changes.

Lab Activities

1. Create GitHub repository
2. Setup coding environment
3. Create class files
4. Implement basic features

Deliverables

1. GitHub repository link
2. IDE screenshots
3. Sample source code

Example LAB SESSION 06: Implementation – Part 1

1. GitHub Repository

- Link: `https://github.com/SampleStudent/SmartLibraryProject`

2. Sample Source Code (Book.java)

Java

```
public class Book {
    private String isbn;
    private String title;
    private boolean isAvailable;

    public Book(String isbn, String title) {
        this.isbn = isbn;
        this.title = title;
        this.isAvailable = true;
    }

    public void checkOut() {
        if (isAvailable) {
            this.isAvailable = false;
        } else {
            System.out.println("Book is already issued.");
        }
    }
}
```

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LAB SESSION 07

Improving Code Quality

Objective

Improve existing code using good design principles.

Basic Theory

SOLID principles help write better code.

Lab Activities

1. Review code written earlier
2. Identify one problem in design
3. Fix the problem
4. Do pair programming for 30 minutes

Deliverables

1. Before and after code
2. Short explanation of improvements

Example LAB SESSION 07: Improving Code Quality

1. Identified Problem (Code Smell)

- **Violation:** Single Responsibility Principle (SRP).
- **Issue:** The `Book` class was handling book details *and* printing fine receipts to the console.

2. Before Refactoring (Bad Code)

Java
```
public class Book {
    public void calculateFine(int days) {
        int fine = days * 10;
        System.out.println("Printing Receipt: " + fine); // Violation: Book
shouldn't handle printing
    }
}
```

3. After Refactoring (Clean Code)

Java
```
// Logic stays in Book or FineCalculator
public class FineCalculator {
    public int calculate(int days) {
        return days * 10;
    }
}

// Printing moves to a separate service
public class ReceiptPrinter {
    public void print(int amount) {
        System.out.println("Printing Receipt: " + amount);
    }
}
```

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LAB SESSION 08

Unit Testing

Objective

Check whether individual functions work correctly.

Basic Theory

Unit testing tests small parts of code.
Each test checks one condition.

Lab Activities

1. Setup testing framework
2. Write minimum 5 unit tests
3. Include one failure test

Deliverables

1. Test source code
2. Screenshot of test results

Example LAB SESSION 08: Unit Testing

1. Test Framework: JUnit (Java) or PyTest (Python)

2. Unit Test Code Sample

Java

```
import static org.junit.Assert.assertEquals;
import org.junit.Test;

public class FineTest {
    @Test
    public void testFineCalculation() {
        FineCalculator fc = new FineCalculator();
        int result = fc.calculate(5); // 5 days late
        assertEquals(50, result); // Expecting $50 fine
    }

    @Test
    public void testNoFine() {
        FineCalculator fc = new FineCalculator();
        int result = fc.calculate(0); // On time
        assertEquals(0, result);
    }
}
```

3. Result Screenshot

- [PASS] testFineCalculation
- [PASS] testNoFine
- Total Tests: 2, Passed: 2.

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LAB SESSION 09

System Testing

Objective

Test the complete system.

Basic Theory

Integration testing checks interaction between modules.
System testing checks full software behaviour.

Lab Activities

1. Write one integration test
2. Exchange project with another group
3. Test their system
4. Record bugs

Deliverables

1. Integration test report
2. Bug report

Example LAB SESSION 09: System Testing

1. Integration Test Scenario

- **Modules:** `BorrowBook` service + `Database`.
- **Action:** User clicks "Borrow".
- **Check:** Does the database status actually change from "Available" to "Issued"?

2. Bug Report (From Peer Review) | Bug ID | Description | Severity | Status | | :--- | :--- | :--- | :--- | | BUG-01 | Login page crashes if password field is empty. | High | Open | | BUG-02 | "Return Date" allows selecting past dates. | Medium | Open | | BUG-03 | Spelling mistake on Welcome Screen ("Librarry"). | Low | Fixed |

LAB SESSION 10

Final Submission

Objective

Prepare final documentation and presentation.

Basic Theory

Documentation explains how to use and understand the system.

Lab Activities

1. Prepare technical documentation
2. Prepare user manual
3. Create presentation slides
4. Write team reflection

Deliverables

1. Final project report
2. User manual
3. Presentation slides
4. Retrospective note

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Example LAB SESSION 10: Final Submission

1. User Manual Snippet

How to Renew a Book:

1. Log in to your dashboard.
2. Click on "My Books".
3. Click the "Renew" button next to the book title. *Note: You cannot renew if the book is already overdue.*

2. Retrospective (Reflection)

- **What went well:** The team communicated well using Trello; the search feature is fast.
- **What didn't go well:** We struggled with connecting the database initially; testing was rushed at the end.
- **Learning:** Learned how to use Git branches and handle merge conflicts.

3. Final Presentation Slides

- Slide 1: Title & Team.
- Slide 2: Problem Statement.
- Slide 3: Technologies Used.
- Slide 4: Key Features (Screenshots).
- Slide 5: Future Enhancements.

Write a final SRS document with all micro level details.